INCH-POUND

MIL-R-82804(OS)
29 December 1989
SUPERSEDING
WS 21846
25 October 1983

MILITARY SPECIFICATION

RESIN, EPOXY, LIQUID

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a liquid epoxy resin, herein referred to as the resin.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8030

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1963 Specific Gravity of Drying Oils, Varnishes, Resins, and Related Materials at 25/25°C

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

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NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the resin shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Material.** The resin shall be a reaction product of bisphenol A and epichlorohydrin.

3.3 **Chemical and physical properties.** Chemical and physical properties of the resin shall conform to the requirements of table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Specific gravity, 25/25°C	1.16	1.17
Viscosity, 25°C, centipoise	11,000	14,000
Weight per epoxy, gram per gram equivalent	187	193
Hydrolyzable chlorine, percent	---	0.2
Moisture, percent	---	0.05

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3.4 **Shelf life.** The material shall have a shelf life of 1 year from date of manufacture when stored at a material bulk temperature not greater than 110°F.

3.4.1 **Shelf life extension.** The shelf life expiration date of an individual lot of material may be extended for additional 12-month periods provided the material, upon retest, conforms to the viscosity, weight per epoxy, and hydrolyzable chlorine requirements of table I.

3.5 **Infrared identification.** The infrared identification produced by the resin shall conform qualitatively to the identification shown in figure 1.

3.6 **Safety.**

3.6.2 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.6.1 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.7 **Workmanship.** Workmanship shall be such that the resin is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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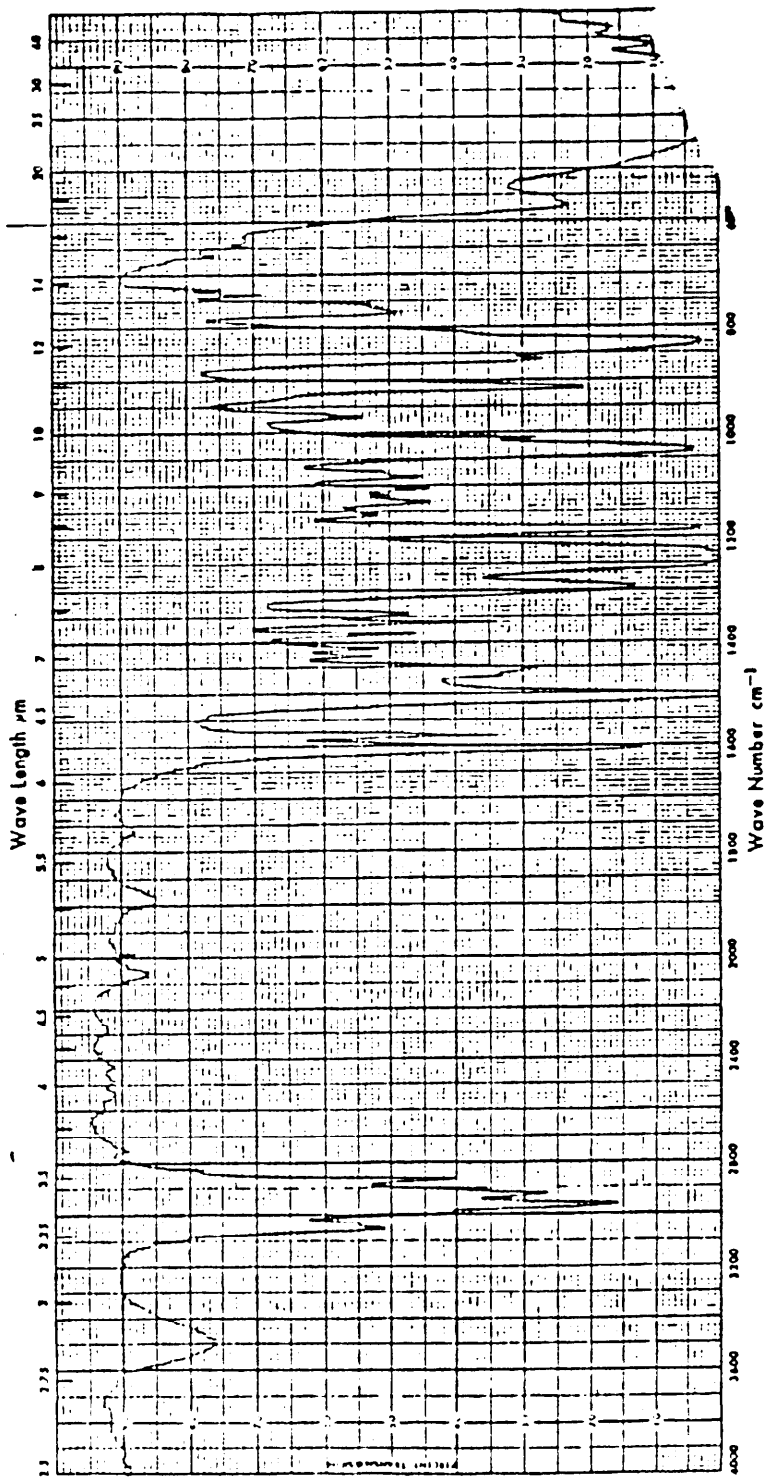


FIGURE 1. Infrared spectrum (DER-331).

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4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

TABLE II. Quality conformance inspection.

Examinations and Tests	Requirement	Method
Visual examination	3.7 and Section 5	4.5.1
Specific gravity	3.3	4.5.2
Viscosity	3.3	4.5.3
Weight per epoxy	3.3	4.5.4
Hydrolyzable chlorine	3.3	4.5.5
Moisture	3.3	4.5.6
Infrared identification	3.5	4.5.7

4.4.1 **Lot.** A lot shall consist of all the liquid epoxy resin offered for acceptance at one time and manufactured by identical processes from one of the following:

a. A single batch (see 6.3.1).

b. A continuous process (see 6.3.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

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4.4.2 **Sampling.** The number of containers to be selected at random from the lot shall be in accordance with table III.

TABLE III. Sampling.

Number of Containers in Lot	Number of Containers to be Selected
1 - 3	All
4 - 40	3

4.4.2.1 **Sample size.** Sample size, selected at random from the lot, shall be a minimum of one pint.

4.4.2.2 **Sample preparation.** The single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites except where a lot consists of less than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of less than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.7 and section 5.

4.5.2 **Specific gravity.** Specific gravity shall be determined in accordance with ASTM D 1963 using a Hubbard pycnometer. Specific gravity shall be recorded to the nearest 0.001 unit.

4.5.3 **Viscosity.** Viscosity shall be determined in accordance with the following:

a. Fill the tube of a Cannon-Manning vacuum viscometer to within $\pm$ 0.5 inch of the "full" line, allowing the sample to run down the side of the tube to prevent air entrapment.

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b. Condition the viscometer and sample in a constant temperature bath at 25.0 ± 0.1°C for a minimum of 30 minutes.

c. Exclude all air from the sample before starting the run.

d. Using a vacuum control capable of controlling to ± 0.1 centimeter (cm), stabilize the regulator at 30 cm of mercury and attach rubber tube from the regulator to the viscometer.

e. Read the efflux time for the small bulb and for the large bulb.

f. Calculate the viscosity as follows:

Viscosity, centipoise = T x C x 100

Where: T = efflux time, seconds

C = calibration factor for bulb, in poises

g. Average the viscosity obtained from each bulb and record to the nearest 100 centipoise.

4.5.4 **Weight per epoxy.** Weight per epoxy shall be determined in accordance with the following:

a. Weigh a 0.3- to 0.4-gram (g) sample to the nearest 0.1 milligram (mg).

b. Dissolve the sample in 10 milliliters (ml) of chlorobenzene and add 5 drops of crystal violet indicator (0.1 percent in glacial acetic acid).

c. Titrate the solution to a permanent blue-green endpoint with 0.1 normal (N) standard hydrobromic acid (HBr) in glacial acetic acid.

d. Calculate the weight per epoxy as follows:

$$\text{Weight per epoxy} = \frac{1000\ W}{VN}$$

Where: W = sample weight, g

V = volume of standard HBr, ml

N = normality of standard HBr

e. Record the weight per epoxy to the nearest gram per gram-equivalent.

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4.5.5 **Hydrolyzable chlorine.** Hydrolyzable chlorine shall be determined in accordance with the following:

a. Weigh a 8.0- to 9.0-gram sample to the nearest 0.1 mg.

b. Dissolve the sample in 50.0 ml of 0.1N alcoholic potassium hydroxide and approximately 15 ml of toluene.

c. Gently reflux the solution for 15 +2, -0 minutes and allow to cool to room temperature.

d. Titrate the solution potentiometrically or to a phenophthalein endpoint with 0.1N standard hydrochloric acid (HCl).

e. Carry a reagent blank through the entire procedure.

f. Calculate the hydrolyzable chlorine as follows:

$$\text{Percent hydrolyzable chlorine} = \frac{3.55\ (B - A)\ N}{W}$$

Where: B = volume of standard HCl for the blank, ml

A = volume of standard HCl for the sample, ml

N = normality of standard HCl

W = standard weight, g

g. Record the hydrolyzable chlorine to the nearest 0.01 percent.

4.5.6 **Moisture.** Moisture shall be determined in accordance with the following:

a. Weigh a 6.0-g sample to the nearest 0.01 g.

b. Dissolve the sample in approximately 100 ml of 3 to 1 pyridine-methanol carrier solution which has been neutralized with Karl Fischer (KF) reagent.

c. Titrate the moisture using an automatic titrator with standardized dilute KF reagent (prepared by diluting commercial KF reagent with pyridine or KF diluent to a water equivalent of 1.8 to 2.3 mg of water per ml of KF reagent).

d. Consider the determination complete when an endpoint is reached which is stable for 30 seconds.

e. Calculate the total moisture as follows:

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$$\text{Percent moisture} = \frac{VE}{10W}$$

Where V = volume of standardized KF reagent, ml

E = water equivalent of standardized KF reagent, mg/ml

W = weight of sample, g

f. Record total moisture to nearest 0.01 percent.

4.5.7 **Infrared identification.** Infrared spectrum shall be determined using an infrared spectrophotometer in accordance with the following:

a. Smear a sample onto a sodium chloride infrared window.

b. Rapidly scan smear between 5 and 15 micrometers (2000 and 650 reciprocal centimeters) without tracing the spectrum. Adjust the smear thickness so the major absorption peak(s) are between 0 and 30 percent transmission range and the background (baseline) is between 80 and 100 percent transmission range.

c. Trace the spectrum from 2.5 to 15 micrometers (4000 to 650 reciprocal centimeters) using instrument settings recommended by the instrument manufacturer for a routine scan.

d. Compare the spectrum obtained from figure 1 and report any deviations.

4.5.8 **Storage life extension.** When required, the tests for storage life extension specified in 3.4.1 shall be in accordance with 4.5.3, 4.5.4, and 4.5.5.

4.6 **Inspection of packaging.** The resin containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The resin shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

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5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Net weight of contents.

d. Shelf life and inspection/test date for extension (see 3.4.1). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTHS PROVIDED THE RESIN, UPON RETEST, CONFORMS TO THE VISCOSITY, WEIGHT PER EPOXY, AND HYDROLYZABLE CHLORINE REQUIREMENTS OF MIL-R-82804."

e. Storage conditions.

f. Toxicity warning (if applicable).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The resin is intended for use in the manufacture of liquid epoxy sealants.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article inspection is required (see 3.1 and 6.4).

d. Material Safety Data Sheets Required (see 3.6.1 and 6.6).

e. Special marking, if other than as specified in 5.2.2.

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6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of resin made as one unit in the manufacturing process.

6.3.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the resin is accumulated for the lot.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Possible material.** The following material has been found to meet the requirements of this specification. This is given only for information and is not restrictive. DER-331 manufactured by Dow Chemical Company has been found to meet the requirements of this specification.

6.6 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 8030-N111)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-82804	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
RESIN, EPOXY, LIQUID

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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